

Kavinda Kehelella

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Research Interests

Natural Language Processing, Large Language Models, LLM Robustness & Consistency, Knowledge Distillation, Domain-Adapted Language Models, Machine Learning, Signal Processing

Education

University of Maryland Aug. 2023 – Present
Ph.D. in Electrical and Computer Engineering (CGPA: 3.88/4.00)

University of Moratuwa Aug. 2017 – May 2022
B.Sc. Engineering (Hons.), Electronic and Telecommunication Engineering, First Class (CGPA: 3.99/4.2)

Dean's List for 7 semesters

Research Experience

Counterfactual Distillation for Robust Question Answering 2026

- Developed a knowledge-distillation framework that selects counterfactual training examples (via entity substitution) to improve student-model robustness to context-vs-parametric knowledge conflicts in open-domain QA.
- Built a T5 teacher-student distillation pipeline (t5-large → t5-base) evaluated on MRQA-style extractive QA (Natural Questions, HotpotQA), reporting exact-match/F1 and teacher-student agreement rates alongside a theoretical Hausdorff-distance bound on the counterfactual selection criterion.
- Diagnosed and fixed a padding-token masking bug in the KL-divergence distillation loss, and ran significance testing and OOD generalization experiments (NarrativeQA, QuAC) in response to reviewer feedback. *Manuscript under review, NeurIPS 2026.*

Improving LLM Consistency Across Paraphrased Inputs 2026

- Investigated why LLMs produce inconsistent outputs on semantically equivalent, paraphrased questions by analyzing hidden-state representations layer-by-layer (representational dissimilarity matrices, PCA) on Gemma-2B-it over HotpotQA.
- Trained probing classifiers to locate the attention heads most responsible for paraphrase-induced inconsistency, then applied targeted activation-editing interventions at those heads to reduce inconsistency.
- Built an evaluation pipeline reporting per-head and per-layer inconsistency deltas before/after editing, with aggregate improvement rates across evaluation instances.

LLM Embeddings for Materials Science Property Classification 2026

- Designed an embedding pipeline for biopolymer formulation classification using domain-specific pretrained language models: ChemBERTa-2 (SMILES structural embeddings) and MatSciBERT (embeddings from functional-group-enriched formulation text).
- Engineered a functional-group-aware text-generation step that encodes per-component chemistry and inter-component interaction rules (e.g., Schiff-base crosslinking, esterification) into natural-language descriptions to better align with MatSciBERT's pretraining distribution.
- Combined structural and semantic embeddings (concatenated 1152-dim representation) to improve downstream classification accuracy over a GNN-only baseline.

RNDC-Lab: Real-Time Calcium Imaging Based Adaptive Neural Decoding May 2024 – Present

- Developing a real-time, end-to-end neural decoding system by integrating two previously developed systems (NDSEP and NeuroGRS) for improved suitability to real-time decoding, using simulated and artificially generated calcium imaging datasets.

Dynamically Reconfigurable Perception via Dataflow Parameterization May 2024 – Present

- Developed a parameterized pruning method to compress large deep learning models for deployment in resource-constrained environments, pruning at inference time; experiments on ResNet50 with a Convolutional Block Attention Module on ImageNet.

Sentiment Transition Analysis in Updated Amazon Reviews Aug. 2024 – Dec. 2024

- Extracted and analyzed updated Amazon reviews across five product categories, comparing sentiment-analysis approaches (GPT-4, a BERT-based sentiment classifier, VADER, and a fine-tuned DistilBERT), highlighting the strengths and limitations of each.

Publications

- *Counterfactual distillation for robust question answering under knowledge conflicts.* Under Review, NeurIPS 2026.
- K. Kehelella, G. Leelarathne, D. Marasinghe, N. Kariyawasam, V. Ariyaratna, A. Madanayake, R. Rodrigo, C. US Edussooriya, “Vision Transformer with Convolutional Encoder-Decoder for Hand Gesture Recognition using 24 GHz Doppler Radar,” *IEEE Sensors Letters*, 2022.

Work Experience

Software Engineer, MyBudget Holdings (Pvt) Ltd., Colombo, Sri Lanka May 2022 – Aug. 2023

- Designed a customer web portal using Go and Java; integrated a PostgreSQL/Kafka backend; maintained documentation via Confluence.

Intern Software Engineer, Axiata Digital Labs, Colombo, Sri Lanka Oct. 2020 – Apr. 2021

- Backend development with Spring Boot; implemented unit tests and worked within an agile team.

Teaching Experience

Graduate Teaching Assistant, Dept. of ECE, University of Maryland Aug. 2023 – Present
ENEE440 Microprocessors; ENEE245 Digital Circuits and Systems Laboratory; ENEE408C/M Capstone Design Projects

Visiting Instructor, Dept. of Electronic & Telecom. Eng., University of Moratuwa Jul. 2022 – Jul. 2023
EN1094 Laboratory Practice; EN4720 Cyber Physical Systems

Skills

Languages: Python, C++, MATLAB, Java, Go, C

ML/NLP Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, Keras

LLM/NLP Tools: T5, Gemma, ChemBERTa-2, MatSciBERT, BERT/DistilBERT, prompt-based classifiers

Other Tools: Git, Jupyter, SQL (PostgreSQL, MySQL), JIRA, Confluence

Honors and Awards

- Graduate School Dean’s Fellowship, University of Maryland (2023)

- Finalist, Hackstat 2.0 – ML/Data Science Competition, University of Colombo (2019)
- Finalist, SLIIT Datathon – Data Science Competition (2019)
- Finalist, Datastorm 1.0 – ML/Data Science Competition, University of Moratuwa (2019)
- Ranked 243rd (World), IEEE Xtreme 14.0 (2020); Ranked 489th (World), IEEE Xtreme 13.0 (2019)
- Mahapola Merit Scholarship, Government of Sri Lanka (2017 – 2022)